

SIMATS ENGINEERING

ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Precision (Level 3)
Question Count: 50

Total Marks: 100

Code of Conduct

I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Thamaraiselvi S**

Assessment Tasks

MCQ Test

1.A company wants to run several virtual machines on a single physical server to reduce hardware costs. Which technology should it use?

- A) Encryption
- B) Virtualization
- C) Authentication
- D) Compression

2.A student wants to access a cloud-based document application from a laptop without installing the application. Which component is most suitable?

- A) Web browser
- B) CPU
- C) RAM
- D) Storage array

3.A hospital stores patient records in the cloud and wants to prevent unauthorized users from reading the data. Which technique should be used?

- A) Encryption
- B) Load balancing
- C) Caching
- D) Compression

4. An organization experiences heavy traffic on its cloud application and wants to distribute requests across several servers. Which technology should it implement?

- A) Firewall
- B) Load balancer
- C) Web browser
- D) Storage array

5. A startup wants to rent virtual machines, storage, and networking resources instead of purchasing physical infrastructure. Which cloud service model is appropriate?

- A) SaaS
- B) PaaS
- C) IaaS
- D) API

6. A user wants to access an online cloud storage service from a desktop computer. What does the user primarily require?

- A) Web browser and Internet connection
- B) Printer and scanner
- C) BIOS and CPU
- D) Keyboard and mouse only

7. A software company wants an environment containing development tools and runtime services to build and deploy applications. Which model should it select?

- A) IaaS
- B) PaaS
- C) SaaS
- D) SAN

8. A teacher uses an online application to create, edit, and share documents with students without installing desktop software. What type of application is being used?

- A) Web application
- B) Device driver
- C) Operating system
- D) Firmware

9. An e-commerce application needs to obtain payment information from an external payment service. Which technology can facilitate this communication?

- A) Web API
- B) RAM
- C) BIOS
- D) Monitor

10. A user enters a cloud application's web address into Chrome and accesses the application. What role does Chrome perform?

- A) Cloud server
- B) Cloud client
- C) Storage array
- D) Database server

11. A company wants employees to access files from different offices without maintaining local storage servers at every office. Which solution is most appropriate?

- A) Cloud storage
- B) Local RAM
- C) BIOS storage
- D) CPU cache

12. A bank requires high-speed block-level storage that can be shared by multiple servers. Which technology should it use?

- A) SAN
- B) Web browser
- C) SaaS
- D) HTML

13. A data center is designing a SAN and needs a centralized device that provides storage capacity to connected servers. Which component is required?

- A) Storage array
- B) Web browser
- C) Keyboard
- D) Application server only

14. Five students located in different cities need to work simultaneously on the same project document. Which cloud feature would help them most?

- A) Cloud collaboration
- B) Local-only storage
- C) BIOS configuration
- D) CPU scheduling

15. An organization wants applications from different cloud providers to communicate effectively. What should it emphasize?

- A) Cloud standards
- B) Hardware replacement
- C) Screen resolution
- D) Local storage

16. A business experiences unpredictable demand during festival seasons. It wants resources to increase during peak periods and decrease afterward. Which cloud characteristic is most useful?

- A) Elasticity
- B) Manual processing
- C) Local storage
- D) Fixed capacity

17. A company subscribes to an online email application and does not manage the underlying servers or operating system. Which model is being used?

- A) IaaS
- B) PaaS
- C) SaaS
- D) SAN

18. A developer wants to deploy an application without managing physical servers or operating systems. Which cloud model is most suitable?

- A) PaaS
- B) SAN
- C) LAN
- D) DNS

19. A company wants only authorized employees to access its cloud application. Which security mechanism should be implemented?

- A) Authentication
- B) Compression
- C) Caching
- D) Load balancing

20. After logging into a cloud system, an employee should only be allowed to access financial reports. Which security process determines what the employee can access?

- A) Authorization
- B) Encryption
- C) Compression
- D) Routing

21. A company wants to block unauthorized network traffic entering its cloud infrastructure. Which security component should it deploy?

- A) Firewall
- B) Web browser
- C) Storage array
- D) CPU

22. A bank's customers access their accounts through a website. The bank wants communication between customers and the server to be encrypted. Which protocol should be used?

- A) HTTP
- B) HTTPS
- C) FTP
- D) SMTP

23. A mobile application retrieves weather information from an external cloud service. Which technology is most likely used?

- A) Web API
- B) RAM
- C) BIOS
- D) SAN

24. An organization develops a mobile application that communicates with a cloud server using HTTP requests. What is being used?

- A) Web API
- B) Storage array
- C) CPU cache
- D) Virtual memory

25. A company wants employees to access business files from their office, home, and mobile devices. Which solution is most appropriate?

- A) Cloud storage
- B) Local hard disk only
- C) Offline storage
- D) BIOS

26. An online business expects its data volume to grow rapidly. Which cloud storage advantage would be most useful?

- A) Scalability
- B) Fixed storage capacity
- C) Manual backup only
- D) Local-only access

27. A video processing company needs very fast access to large amounts of data from multiple servers. Which solution is suitable?

- A) SAN
- B) Web browser
- C) SaaS
- D) Email

28. A project team wants members to edit a shared presentation and see updates immediately. Which capability should they use?

- A) Real-time cloud collaboration
- B) Local file transfer
- C) BIOS configuration
- D) Offline backup

29. A company wants to migrate its application between different cloud providers without major compatibility problems. What is important?

- A) Interoperability standards
- B) Screen resolution
- C) Keyboard type
- D) Local printer

30. A small company cannot afford to purchase expensive servers but needs computing resources for its new application. What is a major advantage of cloud computing for this company?

- A) Pay-as-you-use resource availability
- B) Mandatory hardware purchase
- C) Permanent fixed capacity
- D) No Internet access

31. A developer creates an application that automatically increases the number of instances when user traffic becomes high. Which cloud concept is being applied?

- A) Auto-scaling
- B) Encryption
- C) Authentication
- D) Data compression

32. A development team wants its application to be scalable, resilient, and optimized for cloud infrastructure. Which approach should it follow?

- A) Cloud-native development
- B) Hardware-only development
- C) Offline development
- D) Manual deployment only

33. A large application is difficult to maintain because all functionality is contained in one large program. The development team wants to divide it into smaller independently deployable components. Which architecture should it choose?

- A) Microservices
- B) SAN
- C) Monolithic storage
- D) BIOS architecture

34. A government organization wants its cloud infrastructure to be used exclusively by its departments. Which deployment model is appropriate?

- A) Private cloud
- B) Public cloud
- C) Hybrid cloud
- D) Community website

35. A company stores sensitive financial information in its private cloud but uses a public cloud for its website during high-traffic periods. Which model is being used?

- A) Private cloud
- B) Public cloud
- C) Hybrid cloud
- D) Community cloud

36. A company uses services from two different cloud providers for computing and backup. What type of cloud strategy is this?

- A) Private cloud
- B) Multi-cloud
- C) Single cloud
- D) Local computing

37. A user enters a URL into a browser, and the browser sends a request to a cloud server. What is the browser acting as?

- A) Client
- B) Database
- C) Storage array
- D) Firewall

38.

An online shopping application must remain available even when one server fails. What approach should the company use?

- A) Multiple servers with redundancy
- B) One server only
- C) Local storage only
- D) Manual data entry

39. A researcher stores confidential research data in cloud storage and wants to protect it even if unauthorized people obtain the storage files. What should be implemented?

- A) Encryption
- B) Load balancing
- C) Auto-scaling
- D) Caching

40. A company wants to detect whether cloud-stored files have been modified without authorization. Which mechanism is useful?

- A) Hashing
- B) Load balancing
- C) Virtualization
- D) Scaling

41. A university wants students to use an online learning management application without installing or maintaining the software. Which service model is most appropriate?

- A) SaaS
- B) IaaS
- C) PaaS
- D) SAN

42. A developer has completed a cloud application and wants to identify errors before releasing it to users. What should be performed?

- A) Application testing
- B) Hardware replacement
- C) Data deletion
- D) Network disconnection

43. A company wants to recover important files after accidental deletion or system failure. Which service should it implement?

- A) Cloud backup
- B) Web browser
- C) Load balancing
- D) API gateway only

44. A cloud administrator wants to identify which virtual machines are consuming excessive CPU resources. Which activity is required?

- A) Resource monitoring
- B) Data encryption
- C) Web browsing
- D) File compression

45. A company wants cloud resources to be automatically created whenever application demand increases. Which approach is most suitable?

- A) Cloud automation
- B) Manual configuration
- C) Local-only storage
- D) Offline processing

46. A company wants to use an online CRM system without purchasing servers or installing the application on employee computers. Which service is suitable?

- A) SaaS
- B) SAN
- C) IaaS only
- D) LAN

47. A cloud application becomes slow because too many users are accessing a single server. Which solution would improve traffic distribution?

- A) Load balancing
- B) Data encryption
- C) Authentication
- D) File compression

48. A financial application uses an API to exchange sensitive customer information. Which practice is most important?

- A) Use secure HTTPS communication and authentication
- B) Disable authentication
- C) Use unencrypted HTTP only
- D) Allow unrestricted access

49. A development team wants to deploy an application quickly without manually configuring every server. Which cloud capability would help?

- A) Automated deployment
- B) Manual installation
- C) Local-only execution
- D) Hardware replacement

50. A company wants to develop a web application, store its data in the cloud, provide access through browsers, communicate with external services using APIs, and automatically scale during high demand. Which approach best satisfies these requirements?

- A) Cloud-based application architecture
- B) Standalone desktop architecture
- C) Local-only storage architecture
- D) Offline computing architecture

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Answers with Reasons

Q.No	Answer	Reason
1	B) Virtualization	Virtualization allows multiple virtual machines to run on a single physical server, reducing hardware cost.
2	A) Web browser	A cloud-based document application can be accessed through a browser without installing the application locally.
3	A) Encryption	Encryption converts data into an unreadable form so unauthorized users cannot understand the patient records.
4	B) Load balancer	A load balancer distributes incoming requests across multiple servers to handle heavy traffic.
5	C) IaaS	IaaS provides virtual machines, storage and networking as cloud infrastructure services.
6	A) Web browser and Internet connection	Cloud storage is accessed through the Internet, commonly using a web browser.
7	B) PaaS	PaaS provides development tools, runtime environments and platforms for building and deploying applications.
8	A) Web application	An online application accessed through a browser is a web application.
9	A) Web API	A Web API allows an application to communicate with external services such as payment systems.
10	B) Cloud client	Chrome acts as the client through which the user accesses the cloud application.
11	A) Cloud storage	Cloud storage allows files to be accessed from different offices without maintaining local storage servers.
12	A) SAN	SAN provides high-speed block-level storage that can be accessed by multiple servers.
13	A) Storage array	A storage array provides centralized storage capacity to servers in a SAN environment.
14	A) Cloud collaboration	Cloud collaboration allows users in different locations to work on the same document.
15	A) Cloud standards	Standards improve compatibility and communication between different cloud providers.
16	A) Elasticity	Elasticity allows cloud resources to increase during high demand and decrease when demand falls.

Q.No	Answer	Reason
17	C) SaaS	SaaS provides ready-to-use applications without requiring users to manage servers or operating systems.
18	A) PaaS	PaaS allows developers to deploy applications without managing the underlying physical infrastructure.
19	A) Authentication	Authentication verifies whether a person is an authorized user.
20	A) Authorization	Authorization determines what resources an authenticated user is permitted to access.
21	A) Firewall	A firewall controls and blocks unauthorized network traffic.
22	B) HTTPS	HTTPS encrypts communication between the user's browser and the server.
23	A) Web API	A Web API enables the mobile application to obtain weather information from an external service.
24	A) Web API	HTTP-based communication between an application and cloud server commonly uses a Web API.
25	A) Cloud storage	Cloud storage enables employees to access business files from different devices and locations.
26	A) Scalability	Scalability allows storage capacity to increase as the amount of data grows.
27	A) SAN	SAN provides fast access to large amounts of block-level data for multiple servers.
28	A) Real-time cloud collaboration	It allows team members to edit shared files and see updates immediately.
29	A) Interoperability standards	Interoperability standards help applications move or communicate across different cloud providers.
30	A) Pay-as-you-use resource availability	Cloud computing lets small companies obtain computing resources without purchasing expensive physical servers.
31	A) Auto-scaling	Auto-scaling automatically increases or decreases application instances according to demand.
32	A) Cloud-native development	Cloud-native applications are designed for scalability, resilience and efficient cloud operation.
33	A) Microservices	Microservices divide a large application into smaller independently deployable services.
34	A) Private cloud	A private cloud is dedicated to one organization or group.

Q.No	Answer	Reason
35	C) Hybrid cloud	Combining private-cloud resources with public-cloud resources is a hybrid cloud approach.
36	B) Multi-cloud	Using services from two different cloud providers is called a multi-cloud strategy.
37	A) Client	The browser sends requests to the cloud server, so it acts as the client.
38	A) Multiple servers with redundancy	Redundant servers allow the application to remain available when one server fails.
39	A) Encryption	Encryption protects confidential data even if unauthorized users obtain the stored files.
40	A) Hashing	Hashing can help detect whether stored data has been changed by comparing hash values.
41	A) SaaS	SaaS provides ready-to-use applications without requiring users to install or maintain the software.
42	A) Application testing	Testing identifies errors and problems before the application is released.
43	A) Cloud backup	Cloud backup helps recover files after accidental deletion or system failure.
44	A) Resource monitoring	Monitoring identifies which virtual machines are consuming excessive CPU and other resources.
45	A) Cloud automation	Automation can automatically create and configure resources when demand increases.
46	A) SaaS	An online CRM is a ready-to-use software service delivered through the cloud.
47	A) Load balancing	Load balancing distributes requests across servers and prevents one server from becoming overloaded.
48	A) Use secure HTTPS communication and authentication	HTTPS protects data during transmission and authentication restricts access to authorized users.
49	A) Automated deployment	Automated deployment reduces manual server configuration and speeds up application deployment.
50	A) Cloud-based application architecture	It supports web access, cloud storage, APIs and automatic scaling, satisfying all the stated requirements.